Lift Right — PACT Analysis

CET257 Enterprise Project — Assignment 1, Task A1 — Team Speed

1.Why PACT

PACT (People, Activities, Contexts, Technologies) was chosen over a purely feature-driven requirements pass because Lift Right's biggest design risks are not technical, they are situational: a shared warehouse PC, a worker who has never used a first-person control scheme, and a five-minute training slot squeezed between shifts. PACT forces those conditions to be written down before any interface or checkpoint content is designed, so decisions are traceable back to a real constraint rather than to what looked good in a mock-up. Each section below ends with the design consequence the team drew from it, and Section 6 lists the concrete build decisions that trace back to this analysis.

2. People

Attribute	What we found	Design consequence
Warehouse operatives	Wide range of ages, physical fitness and prior digital/gaming experience; some are new starters on their first day.	Controls limited to look (mouse) and W-A-S-D movement, with a short pre-start briefing screen rather than an in-depth tutorial.
Non-native English speakers	The floor is expected to include workers for whom English is a second language.	Checkpoint prompts and quiz questions are written as short, plain sentences and paired with clear visual cues (highlighted load, marker colour) rather than relying on text alone.
Supervisors / training administrators	Time-poor, need a fast answer to "who is trained", not a technical audience and not expected to read logs.	Admin Command Center is a single summarised table (logins, attempts, time-on-task, best score) with a one-click CSV export rather than a raw data dump.

3. Activities

Attribute	What we found	Design consequence
Completing training	Typically one short sitting of five to ten minutes, often fitted around shift start or a quiet moment on the floor.	The six-checkpoint run was scoped to be completable in one sitting; progress is saved so an interrupted session is not lost.

Making safety decisions under pressure	Real lifting judgement calls (check the route, assess the load, pick a method) happen quickly, not with time to consult a manual.	Each checkpoint presents one decision at a time with a visible countdown, mirroring the pace of a real judgement call rather than an open-book quiz.
Reviewing competency records	Supervisors check who has completed training periodically, ahead of audits, or after an incident, rather than continuously.	Records persist between sessions and can be exported on demand, rather than requiring the supervisor to be present while training happens.

4. Contexts

Attribute	**What we found**	**Design consequence**
Physical setting	Likely a shared workstation or break-room PC on the warehouse floor, possibly with background noise and limited privacy.	No dependency on audio for critical information; all essential feedback (score, teaching card, timer) is shown on screen.
Interruptions	A session may be paused if a worker is called back to the floor mid-attempt.	Login and per-worker progress are stored client-side so a worker's account state survives being interrupted and returning later.
IT support	No dedicated on-site IT support is assumed to install software or maintain a server for the prototype stage.	The prototype was built as a single self-contained browser file requiring no installation, consistent with NFR6 in the Requirements document.

5. Technologies

Technology	**Decision**	**Reason**
Rendering engine	Three.js, procedural geometry rather than large imported 3D asset files.	Keeps the scene lightweight enough to run on modest, shared warehouse PCs rather than requiring gaming-spec hardware.

Controls	Pointer-lock mouse-look plus W-A-S-D keyboard movement; no VR headset assumed.	Matches the standard desktop/laptop, mouse-and-keyboard hardware the client actually has on the warehouse floor.
Data persistence	Client-side storage (accounts, attempt history, best scores) rather than a live backend for the prototype.	No guarantee of an on-site server for a first client trial; a backend is flagged as future work rather than assumed.
Delivery format	Single self-contained browser file, Chrome/Edge/Firefox.	No install or plugin requirement, so the client can trial the build on existing hardware immediately.

6. What PACT Changed

Three concrete build decisions trace directly back to this analysis rather than being arbitrary implementation choices:

1. Controls were fixed as mouse-look plus W-A-S-D early in Week 1, once the People and Technologies analysis ruled out assuming any VR or controller hardware on the warehouse floor.

2. Checkpoint and quiz text were kept deliberately short, and the interface avoids dense paragraphs, in direct response to the People finding on non-native English speakers and varying digital literacy.

3. Client-side data persistence and the CSV export were adopted for this prototype stage specifically because of the Contexts finding that no on-site server could be assumed; the Requirements and System Requirement Analysis documents record this as a scoped decision to be revisited with a real backend in a later iteration, not a permanent architecture choice.